MICROBIAL CONTAMINATION OF FRESH RAW VEGETABLE SAMPLES COLLECTED FROM LOCAL MARKETS OF SULAYMANIYAH CITY

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ABSTRACT

This study focuses on evaluating microbial contamination in fresh raw vegetables obtained from local markets in Sulaymaniyah city. Different vegetable samples were examined using microbiological and parasitological methods to identify possible contaminants. The results showed the presence of bacteria such as E. coli, Klebsiella, and Pseudomonas, which indicates potential health risks for consumers. Antibiotic testing also revealed differences in how these bacteria respond to treatment. These findings emphasize the importance of proper hygiene, washing practices, and monitoring to ensure food safety and reduce the risk of foodborne diseases.

Keywords: Microbial contamination, raw vegetables, bacteria, parasites, food safety, Sulaymaniyah

TABLE OF CONTENTS

ABSTRACT 2
TABLE OF CONTENTS 3
LIST OF FIGURES 4
LIST OF TABLES 5
LIST OF ABBREVIATION 6
ACKNOWLEDGMENTS ..... **Error! Bookmark not defined.**
1.1INTRODUCTION 8
1.2 AIM 8
2.1 Literature review 10
2.2 Parasitic Contamination and Microbiological Quality of Fresh Vegetables in Debre Berhane, Ethiopia 10
2.3 Raw Vegetables Sold in Traditional Markets: Contamination Risks & Preventive Measures 11
2.3 Detecting Intestinal Parasites in UAE Vegetables 12
3.1 Materials used 14
3.2 Microbial Contamination 14
3.2.1 Sample processing 14
3.2.2 Parasite detection. 15
3.2.3 Total Coliform Count 15
3.2.4 Bacteria isolation and identification 16
References 22

LIST OF FIGURES

Figure 1 results for biochemical test for bacteria identification 18

LIST OF TABLES

Table 1 effect of antibiotic on bacteria: 19

LIST OF ABBREVIATION

ABREVIATIONS	DEFENETINO
MPN	Most Probable Number
MAC	MacConkey Agar
SSA	Salmonella-Shigella Agar
MSA	Mannitol Salt Agar
TSIA	Triple Sugar Iron Agar

CHAPTER ONE
INTRODUCTION

1.1 INTRODUCTION

Fresh raw vegetables are an essential part of a healthy diet as they are rich in vitamins, minerals,

and fiber. However, they can also be a source of microbial contamination, which poses a significant threat to public health [1]. Bacteria and intestinal parasites are among the most common contaminants found in fresh raw vegetables, and their presence can cause foodborne illnesses [2].

In Sulaymaniyah city, Iraq, fresh raw vegetables are commonly sold in local markets, and them safety regarding microbial contamination is not adequately monitored. The risk of foodborne illness due to consuming contaminated vegetables is high, especially for vulnerable populations such as children, pregnant women, and the elderly [3].

The current study aims to determine the types and levels of bacteria and intestinal parasites that contaminate fresh raw vegetables sold in local markets in Sulaymaniyah city. Additionally, the study aims to determine the antibacterial susceptibility of bacterial isolates, which can help in identifying appropriate treatment options in case of foodborne illness [4].

The findings of this study will provide critical insights into the safety of fresh raw vegetables sold in local markets in Sulaymaniyah city and help in developing appropriate strategies to prevent and control microbial contamination [5]. Ultimately, this research can contribute to safeguarding public health and promoting safe consumption of fresh raw vegetables in the local community [6]

1.2 AIM

The current study aimed to determine bacteria and intestinal parasites that contaminate fresh raw vegetables in local markets and determined the antibacterial susceptibility of bacterial isolates.

CHAPTER TWO
LITERATUR REVIEW

2.1 Literature review

2.2 Parasitic Contamination and Microbiological Quality of Fresh Vegetables in Debre Berhane, Ethiopia

The study conducted a cross-sectional assessment of vegetable contamination, considering factors such as parasite species, weather conditions, socio-cultural conditions, sampling season, and analyzed vegetable products. The research aimed to identify associations between parasite contamination and various factors during the marketing period.

A total of 180 vegetable samples were purchased from vendors and analyzed for parasites and microbial load. The study found that out of the 180 samples, 129 parasites were identified, resulting in an overall contamination rate of 41.7%. Protozoa and helminths were the two types of parasites identified, with helminths being more prevalent. About 21.1% of the samples were contaminated with multiple parasite species.

Significant associations were observed between parasite contamination and factors such as the type of produce, finger nail status of vendors/sellers, display medium, type of market, and whether the vegetables were washed prior to display.

The study also revealed higher microbial loads in vegetables during the afternoon compared to the morning, as indicated by total heterotrophic count, total coliform count, fecal coliform count, yeast count, and mold count

To mitigate health risks, the study suggests implementing strategies to reduce contamination, including encouraging specific display mediums and promoting the washing of vegetables before they are displayed for sale.

2.3 Raw Vegetables Sold in Traditional Markets: Contamination Risks & Preventive Measures

The objective of the study was to identify bacteria and intestinal parasites present in raw vegetables sold in traditional markets, as they can contribute to the transmission of food-borne diseases. Raw vegetables, including lettuce, tomato, cabbage, basil, long bean, and cucumber, were randomly selected from eight traditional markets for sampling. Bacterial identification was carried out through culture and microbiological tests, while intestinal parasites were identified using sedimentation and floatation methods. The findings revealed that all of the raw vegetables were contaminated with Escherichia coli (E. coli) (91%), Staphylococcus aureus (84%), and Vibrio cholera (79%). Additionally, 36% of the samples were contaminated with soil-transmitted helminths (STH), and 27% of the samples were contaminated with intestinal protozoa. Among the vegetables, lettuce had the highest level of contamination with bacteria and intestinal parasites. These results demonstrate that raw vegetables sold in traditional markets are susceptible to bacterial and intestinal parasite contamination, posing a risk for the transmission of food-borne diseases. Therefore, proper handling and hygiene practices for raw vegetables are crucial in preventing such contamination and subsequent health risks.

2.3 Detecting Intestinal Parasites in UAE Vegetables

to assess the level of intestinal parasite contamination in selected vegetables in the UAE.

What they did is 218 fresh vegetable samples were collected randomly from farms and supermarkets. After washing and centrifugation, the sediment was examined microscopically for parasites.

the fund out that 15.1% (33/218) of the samples showed the presence of protozoa cysts and helminths eggs. The most common parasites detected were Entamoeba complex (30.3%), Entamoeba coli (18.2%), Trichuris trichiura (12.1%), Strongyloides stercoralis (12.1%), Ascaris lumbricoides (9.1%), Endolimax nana and Enterobius vermicularis (6.1% each), and Giardia lamblia and Hymenolepis nana (3.0% each). There was no significant association between the vegetable type and parasite occurrence.

at the end Raw vegetables can serve as a major source of foodborne disease outbreaks and transmission of intestinal parasites. Public education on safe handling of raw vegetables is recommended.

CHAPTER THREE
METHODOLOGY

3.1 Materials used

Materials	Manufacture
1. Different culture media.	----
2. Oven.	----
3. Loop.	----
4. Tube.	China
5. Incubator.	Germany
6. Autoclave.	BUN
7. Incubator.	Germany
8. Centrifuge.	United state

3.2 Microbial Contamination

3.2.1 Sample processing

Approximately 200 g of each vegetable sample was minced with a sterile knife and cutting board, rinsed by soaking in a beaker containing 500ml of saline (0.85% NaCl) for 20minutes, followed by agitation on a shaker for 5minutes for adequate washing. The samples were then removed from the beaker and the wash solution was transferred to separate test tubes for bacteriological and parasitological analysis. The wash solution used for parasitological analysis was incubated for 1 hour to allow the parasite stages to settle properly. The top physiological saline solution was then discarded carefully without shaking. Finally, the remaining sediment mixture was centrifuged (using Gallen Kamp angle head centrifuge cat.no cfb 700 0100 hz50) at 2000G for 5 minutes. Then the final sediment was examined for parasite stages.

3.2.2 Parasite detection.

The sediment was mixed and stained and unstained smears were examined for parasites. For unstained smears, a drop of sediment was placed on a fresh clean slide and a coverslip was placed carefully to avoid air bubbles and flooding. The specimens were then examined under a light microscope using (100×) and (400×) magnification. Smears were prepared as per the unstained smears but Lugol's iodine solution was added [8].

3.2.3 Total Coliform Count

The Most Probable Number (MPN) method was used to detect and enumerate coliform bacteria present in the vegetable samples as described by Rompré et al. [9]

1. Presumptive test Three single strength lactose broth tubes were labeled as "0.1", another 3 tubes "1" and 3 double strength broth tubes "10". Each "10" tube was aseptically inoculated with 10 mL of sample, the "1" tubes were aseptically inoculated with 1 mL of sample using a 1 mL sterile pipette and the "0.1" tubes were inoculated aseptically with 0.1 mL of sample using a sterile pipette. All of the nine-inoculated tubes were incubated at 37°C for 24–48 h.
2. Confirmed test All primary tubes showing any amount of gas or acid within 24–48 h of incubation was submitted to the confirmed phase. Primary tubes (positive) were gently shaken to re-suspend the organism. The EMB agar plate was inoculated with the positive culture with a sterile inoculating loop. The plate was incubated for 24–48 h at 37° C in the inverted position. Completed test to establish definitely the presence of coliform bacteria and to provide quality control data, the completed test was used on all positive confirmed cases. The lactose fermentation broth tube was inoculated with the isolated colony from an agar plate using an inoculated loop. The nutrient agar was streaked with the colony from an agar plate with an inoculated loop. The organism on the nutrient agar was tested for gram stain, and subjected to biochemical tests, and identified as coliform

3.2.4 Bacteria isolation and identification

We used nutrient agar, McConkey agar (MAC), salmonella-shigella agar (SSA), and mannitol-salt agar (MSA) as culture and growth media. A 100 µL sample was inoculated in agar plate with spread plate technique. Then all plate incubated at 37° C during 24 hours. After culturing process, the bacteria were characterized by gram staining, oxidase test, and by their growth in triple sugar iron agar (TSIA). Gram positive bacteria were continuous to test with coagulase test, and gram-negative bacteria were continuous to evaluate with the indole methyl-red Voges Proskauer iterate (IMVIC) test

CHAPTER FORE
RESULT AND DISCUSSION

4.1 results

After doing the practical work and the biochemical test to find out what types of bacteria we have in the media, we have found 3 bacteria: E-coli, Klebsiella and Pseudomonas. This test was just for exactly identifying what bacteria we have cultured. This was just for us to make sure because it is based on the agars that we used for the culturing of the bacteria.

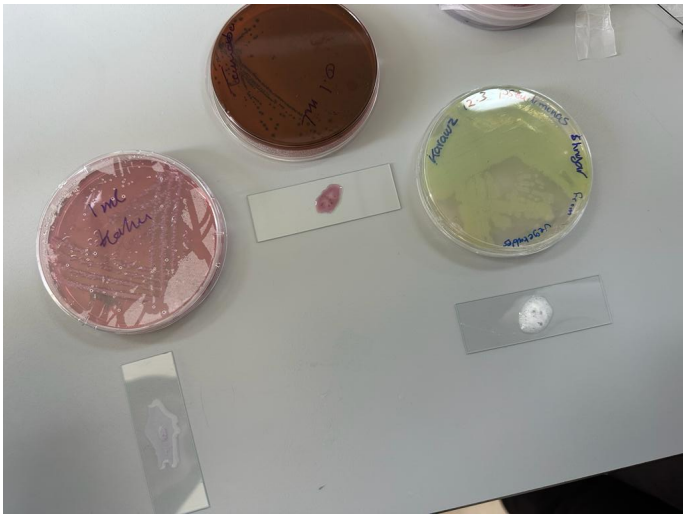

Figure 1 results for biochemical test for bacteria identification

Based on this we used antibiotics for them to see how effective each anti biotic is to the bacteria bum usuring the reads of the effective aria which they are:

1. . etracycline Antibiotics (TE30)
2. IMIPENEM IMI 10 UG (IMI 10)
3. GENTAMICIN GM10(GM10)
4. Fox30 Cefoxitin
5. Penicillin 10

While adding the antibiotics we have got this result of how effective the are on the bacteria

Table 1 effect of antibiotic on bacteria:

Antibiotics	E-coli	pseudomonas	Klebsiella	
			Kawar	lettuce
TE30	2.00 cm	1.50 cm	2.20 cm	No effect
IMI10	2.80 cm	1.00 cm	1.70 cm	No effect
CM10	2.40 cm	2.30 cm	2.30 cm	No effect
FOX30	2.20 cm	No effect	No effect	No effect
P10	No effect	No effect	No effect	No effect

As table (1)shows the results of an experiment to test the effectiveness of different antibiotics against four different types of bacteria: Escherichia coli (E. coli), Pseudomonas aeruginosa, Klebsiella pneumoniae, and Kawar. The experiment was conducted by placing each type of bacteria in a dish with a different antibiotic. The distance that the bacteria grew from the edge of the dish was then measured.

The results of the experiment show that TE30, IMI10, and CM10 were all effective against E. coli, Pseudomonas aeruginosa, and Klebsiella pneumoniae. However, TE30 was not effective against Kawar, and FOX30 and P10 were not effective against any of the bacteria tested.

It is important to note that the results of this experiment are only for the specific strains of bacteria that were tested. The effectiveness of an antibiotic may vary depending on the strain of bacteria. Additionally, the results of this experiment do not necessarily mean that the antibiotics tested are safe to use. More research is needed to determine the safety and effectiveness of antibiotics in humans.

- TE30 was effective against all four types of bacteria, with E. coli being the most sensitive and Kawar being the least sensitive.
- IMI10 was effective against all three Gram-negative bacteria (E. coli, Pseudomonas aeruginosa, and Klebsiella pneumoniae), but not against Kawar.
- CM10 was effective against all three Gram-negative bacteria, but not against Kawar.
- FOX30 was not effective against any of the bacteria tested.
- P10 was not effective against any of the bacteria tested.

The results of this experiment suggest that TE30, IMI10, and CM10 may be effective treatments for infections caused by E. coli, Pseudomonas aeruginosa, and Klebsiella pneumoniae. However, further research is needed to confirm these findings and to determine the safety and effectiveness of these antibiotics in humans.

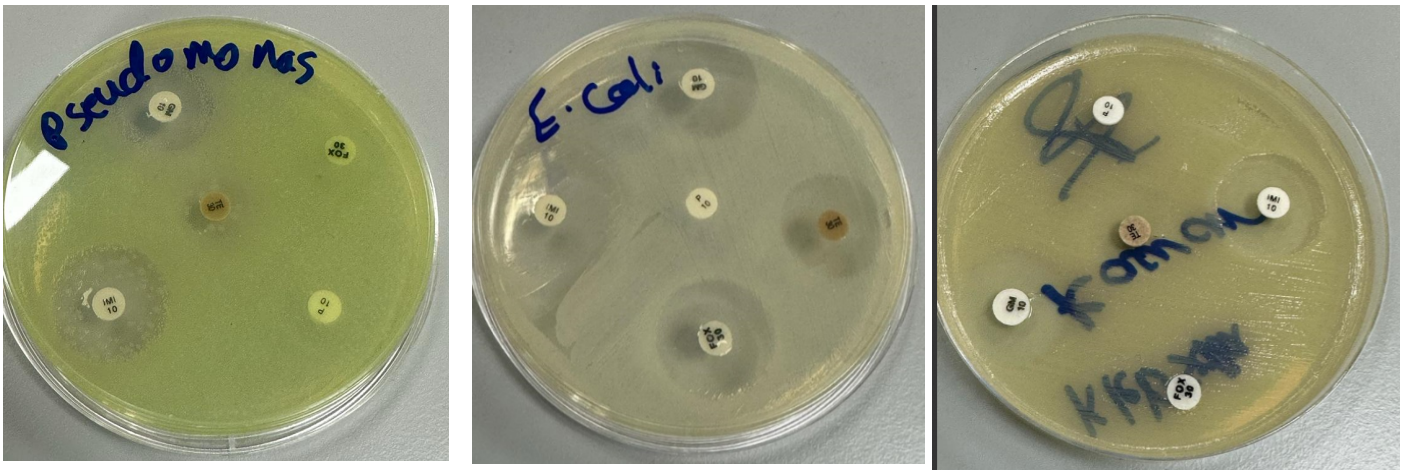

Figure 2 sample of bacteria and the effect of antibiotics

from fig(2) it is

CONCLUSION

The study shows that fresh raw vegetables sold in local markets in Sulaymaniyah city can carry harmful microorganisms. The detection of bacteria such as E. coli, Klebsiella, and Pseudomonas suggests a possible risk when vegetables are consumed without proper cleaning. The variation in antibiotic effectiveness also indicates the need for continuous monitoring of bacterial resistance. Improving hygiene during handling, transportation, and selling of vegetables is essential. In addition, increasing public awareness about proper washing and food safety practices can help reduce the spread of foodborne illnesses.

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